

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomised, Double-Blind, Placebo-Controlled, Multicentre Study Of The Efficacy And Safety Of Atezolizumab Plus Chemotherapy For Patients With Early Relapsing Recurrent (Inoperable Locally Advanced Or Metastatic) Triple-Negative Breast Cancer **Short Title/Acronym:** IMpassion132 **Protocol Number:** M039193 **NCT Number:** NCT03371017 **Posting ID:** 696276685764786027 **Trial Status:** COMPLETED **Sponsor/Company Name:** Hoffmann-La Roche / Roche **Investigational Product:** Atezolizumab plus Chemotherapy (Capecitabine, Gemcitabine [Gemzar, Infugem], or Carboplatin) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy in terms of Overall Survival (OS) of atezolizumab in combination with standard first-line chemotherapy compared to placebo plus chemotherapy in patients with early relapsing recurrent triple-negative breast cancer (TNBC) or Triple Negative Breast Neoplasms. **Secondary Objectives:** To evaluate specific survival rates (12-month and 18-month), Progression-Free Survival (PFS), Objective Response Rate (ORR), Duration of Response (DoR), safety, tolerability, and biomarker evaluations (e.g., PD-L1 status).

2.2 Background & Rationale Triple-negative breast cancer (TNBC) relapsing < 12 months after standard (neo)adjuvant anthracycline and taxane chemotherapy has a notoriously poor prognosis and limited treatment options. While atezolizumab has shown clinical benefit in PD-L1-positive metastatic TNBC (e.g., IMpassion130), this study (IMpassion132) generates critical evidence to determine if adding PD-L1 blockade to standard chemotherapy improves overall survival specifically in the early relapsing, inoperable, locally advanced, or metastatic TNBC setting.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Active Comparator (Chemotherapy) **Blinding:** Double-blind **Randomization:** 1:1 Stratified Randomization

3.2 Planned Enrollment & Duration Target **NS**: 595 patients **Number of Sites**: Global Multicenter **Estimated Study Start**: 11-Jan-2018 **Estimated Primary Completion**: 23-Oct-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically confirmed triple-negative breast cancer (TNBC) that is either locally recurrent, inoperable and cannot be treated with curative intent, or is metastatic. 3. Documented disease progression occurring ≤ 12 months from the last treatment with curative intent (early relapsing). 4. Prior treatment of early breast cancer with an anthracycline and taxane. 5. Have not received prior chemotherapy or targeted systemic therapy for their locally advanced inoperable or metastatic recurrence (prior radiation therapy is permitted). 6. Measurable or non-measurable disease, as defined by RECIST 1.1. 7. Availability of a representative formalin-fixed, paraffin-embedded (FFPE) tumor block (preferred) or at least 17 unstained slides obtained from relapsed metastatic or locally advanced diseases. 8. Eastern Cooperative Oncology Group (ECOG) performance status 0-1. 9. Life expectancy ≥ 12 weeks. 10. Adequate haematologic and end-organ function. 11. Negative tests for HIV, HBsAg, HbCAb (or positive HbCAb followed by negative HBV DNA), and HCV antibody (or positive HCV followed by negative HCV RNA). 12. Agreement to remain abstinent or use a highly effective contraceptive method during the treatment period and for at least 5 months after the last dose of atezolizumab or 6 months after capecitabine, whichever is later. Men must agree to use contraceptive measures and refrain from donating sperm. 13. *For patients enrolled after the recruitment of all-comers is complete*: PD-L1-positive tumor status, defined as PD-L1 expression on tumor-infiltrating immune cells (IC) of 1% or greater.

4.2 Exclusion Criteria 1. Spinal cord compression not definitively treated or clinically stable for > 2 weeks; symptomatic, untreated, or actively progressing central nervous system (CNS) metastases; history of leptomeningeal disease. 2. Symptomatic or rapid visceral progression. 3. No prior treatment with an anthracycline and taxane. 4. Uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures (once monthly or more frequently); uncontrolled tumor-related pain; uncontrolled or symptomatic hypercalcemia. 5. Malignancies other than TNBC within 5 years prior to randomization. 6. Significant cardiovascular disease, unstable arrhythmias, unstable angina, or presence of an abnormal ECG within 3 months prior to randomization. 7. Severe infection requiring oral or IV antibiotics within 4 weeks prior to randomization. 8. Current treatment with anti-viral therapy for HBV. 9. Major surgical procedure within 4 weeks prior to randomization or anticipated need for major surgery during the study. 10. Treatment with investigational therapy within 28 days

prior to randomization. 11. Pregnant or lactating, or intending to become pregnant. 12. *Atezolizumab-specific*: Hypersensitivity/allergy to biopharmaceuticals or components; history of autoimmune disease; prior allogeneic stem cell or solid organ transplantation; history of idiopathic pulmonary fibrosis, drug-induced pneumonitis, or active pneumonitis; active tuberculosis; receipt of a live, attenuated vaccine within 4 weeks; prior treatment with CD137 agonists, anti-PD-1, or anti-PD-L1 therapies; systemic immunostimulatory agents within 4 weeks; systemic corticosteroids or immunosuppressive medications within 2 weeks prior. 13. *Capecitabine-specific*: Inability to swallow pills; malabsorption syndrome or ulcerative colitis; known dihydropyrimidine dehydrogenase (DPD) deficiency. 14. *Carboplatin/Gemcitabine-specific*: Hypersensitivity to platinum-containing compounds or any component of carboplatin or gemcitabine formulations.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patients are randomized 1:1 to receive Atezolizumab 1200 mg or Placebo every 3 weeks (21 days). This is combined with investigator-selected chemotherapy: Capecitabine, or Gemcitabine (including trade names such as Gemzar or Infugem) / Carboplatin. **Route of Administration:** Intravenous (IV) for Atezolizumab/Placebo, Gemcitabine, and Carboplatin; Oral for Capecitabine. **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing chemotherapy-induced toxicities and cancer-related symptoms (e.g., indwelling catheters like PleurX® are allowed for effusions). **Prohibited:** Receipt of a live, attenuated vaccine within 4 weeks prior to randomization; treatment with systemic immunostimulatory agents within 4 weeks prior to randomization; concurrent investigational therapies; ongoing anti-viral therapy for HBV.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor tissue testing (PD-L1 status), Baseline Scans, Viral Serology (HIV/HBV/HCV)
Baseline	Day 0	Randomization, First Dose of Atezolizumab/Placebo + Chemotherapy
Interim	Every 21 Days (\$\pm\$ 3)	Treatment administration, Safety Labs, Efficacy Assessment (Tumor imaging per RECIST 1.1 every 8-9 weeks)
End of Study	Upon Progression	Final Vital Signs, Exit Interview, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Overall Survival (OS) in PD-L1-positive Population:** Defined as time from randomization to death from any cause. Participants without a reported death event at the time of analysis are censored on the date they were last known to be alive (or date of randomization +1 day if no post-baseline data is available).
- 2. OS in Modified Intent-to-treat (mITT) Population:** Defined identically as time from randomization to death from any cause, with corresponding censoring rules applied to the mITT population.

Measurement Method: Kaplan-Meier methodology survival tracking.

7.2 Secondary & Exploratory Endpoints

- 1. 12-month Survival Rate in PD-L1-positive Population**
- 2. 12-month Survival Rate in mITT Population**
- 3. 18-month Survival Rate in PD-L1-positive Population**
- 4. Progression-Free Survival (PFS):** Time from randomization to the first occurrence of disease progression or death, determined per RECIST 1.1.
- 5. Objective Response Rate (ORR) and Duration of Response (DoR):** Evaluated per RECIST 1.1.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs graded per NCI CTCAE via clinical evaluation and laboratory testing at regular study visits.

Serious Adverse Events (SAEs): Must be reported within 24 hours to the sponsor's safety department.

Data Monitoring Committee (DMC): An independent Data Monitoring Committee (iDMC) reviews safety data periodically throughout the study.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Modified Intent-to-Treat (mITT) and PD-L1-positive populations are primary focus for OS analysis; Safety Set for all participants receiving at least one dose of study medication. Stratification by PD-L1 status, visceral metastases, and selected chemotherapy.

Sample Size Justification: Approximately 595 patients to provide sufficient statistical power ($\alpha = 0.05$, Power $\geq 80\%$) to detect a clinically meaningful difference in Overall Survival.

Handling of Missing Data: Time-to-event data (OS, PFS) for patients without an event are right-censored at the last known date they were alive/assessed. If no post-baseline data are available, OS is censored at the date of randomization +1 day. Kaplan-Meier methodology is utilized for median time-to-event estimations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits by Clinical Research Associates (CRAs) to ensure protocol adherence and data integrity. **Audit Trail:** Electronic Case Report Form (eCRF) utilizing comprehensive audit trails, data clarification forms, and centralized data management checks.

11. Study Identifiers & Governance

Primary ID: M039193 **Registry Number:** NCT03371017 **Posting ID:** 696276685764786027 **Trial Status:** COMPLETED **Sponsor/Lead**

Organization: Hoffmann-La Roche / Roche **Study Title:** IMpassion132 (A Study of the Efficacy and Safety of Atezolizumab Plus Chemotherapy for Patients With Early Relapsing Recurrent Triple-Negative Breast Cancer) **Official Regulatory Title:** A Phase III, Randomised, Double-Blind, Placebo-Controlled, Multicentre Study Of The Efficacy And Safety Of Atezolizumab Plus Chemotherapy For Patients With Early Relapsing Recurrent (Inoperable Locally Advanced Or Metastatic) Triple-Negative Breast Cancer

12. Clinical Framework

Primary Condition: Triple Negative Breast Neoplasms (Early Relapsing TNBC) **Diagnosis Threshold:** Recurrence ≤ 12 months from standard (neo)adjuvant anthracycline and taxane chemotherapy; centrally assessed PD-L1 $\geq 1\%$ on IC for selected cohorts. **Medical Methodology:** Systemic Immunotherapy + Chemotherapy combination **Optimization Target:** Overall Survival (including 12- and 18-month rates) and Progression-Free Survival prolongment.

13. Study Procedures & Methodology

Management Pathway: Standard oncologic management with double-blind investigational product administration. **Education Protocol (HCP):** Investigator protocol training, AE management, and RECIST 1.1 tumor assessment training. **Education Protocol (Patient):** Informed consent, treatment schedule education, and contraception counseling. **Quality Audit Mechanism:** Source data verification (SDV) and central data monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or death (approx. 36 months of follow-up post-randomization).

Onsite Visit Cadence: Every 3 weeks for treatment and safety checks. **Remote Monitoring Frequency:** Survival follow-up performed periodically after treatment discontinuation. **Checklist Review Interval:** Tumor imaging every 8-9 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				